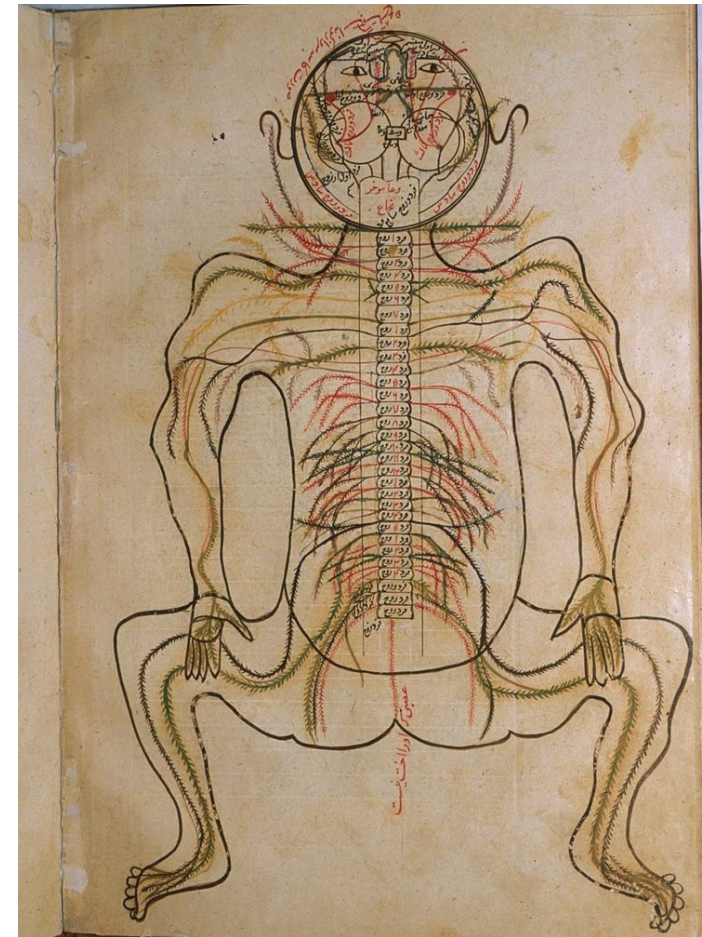


**Specific functions of the  
Autonomic Nervous  
System (ANS):  
Accommodation,  
Micturition, Airways,  
Mucous secretion**

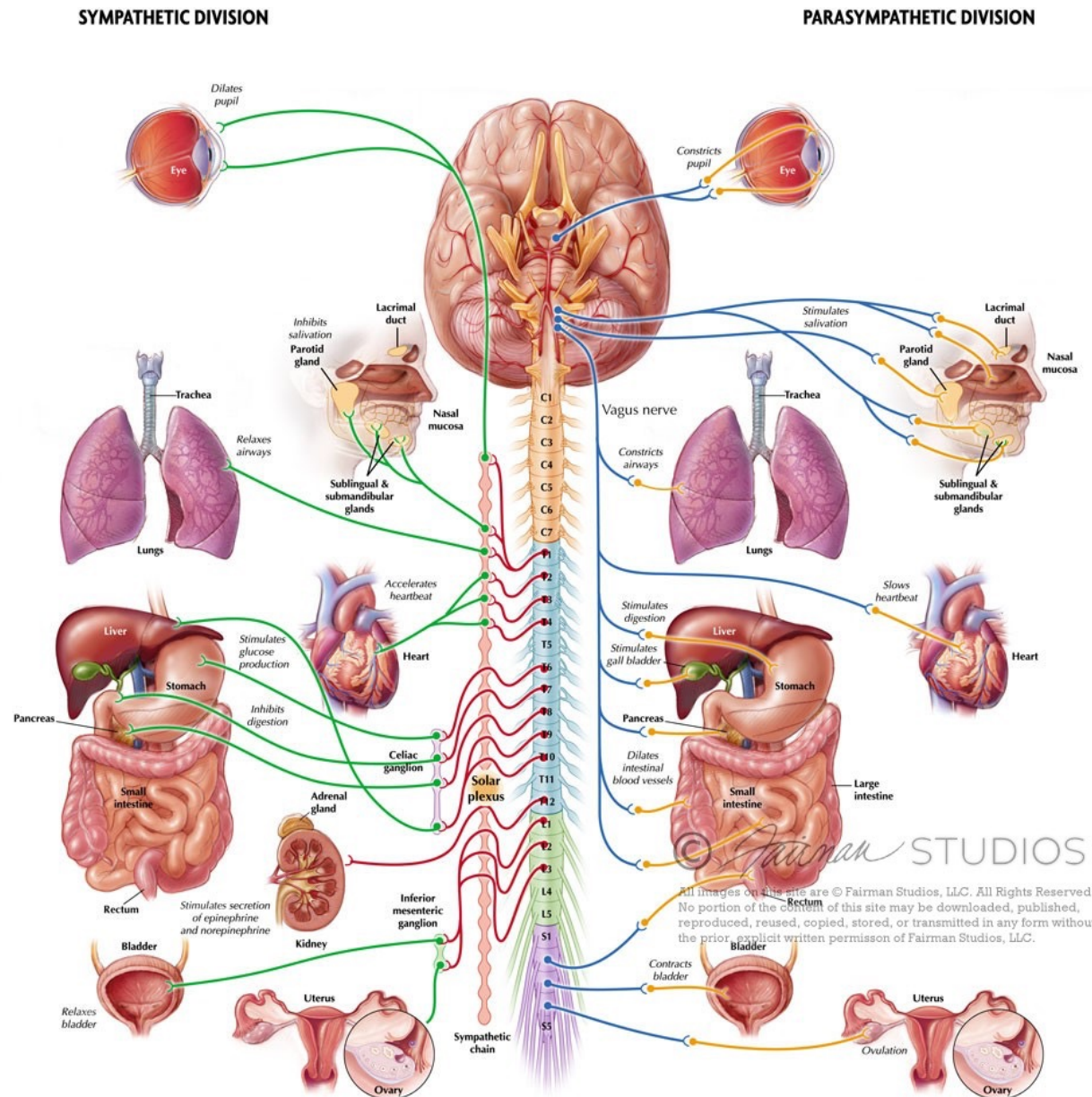


*Anatomical illustration of the nervous system  
by Mansour Ibn-Ilyas, ca. 1400*



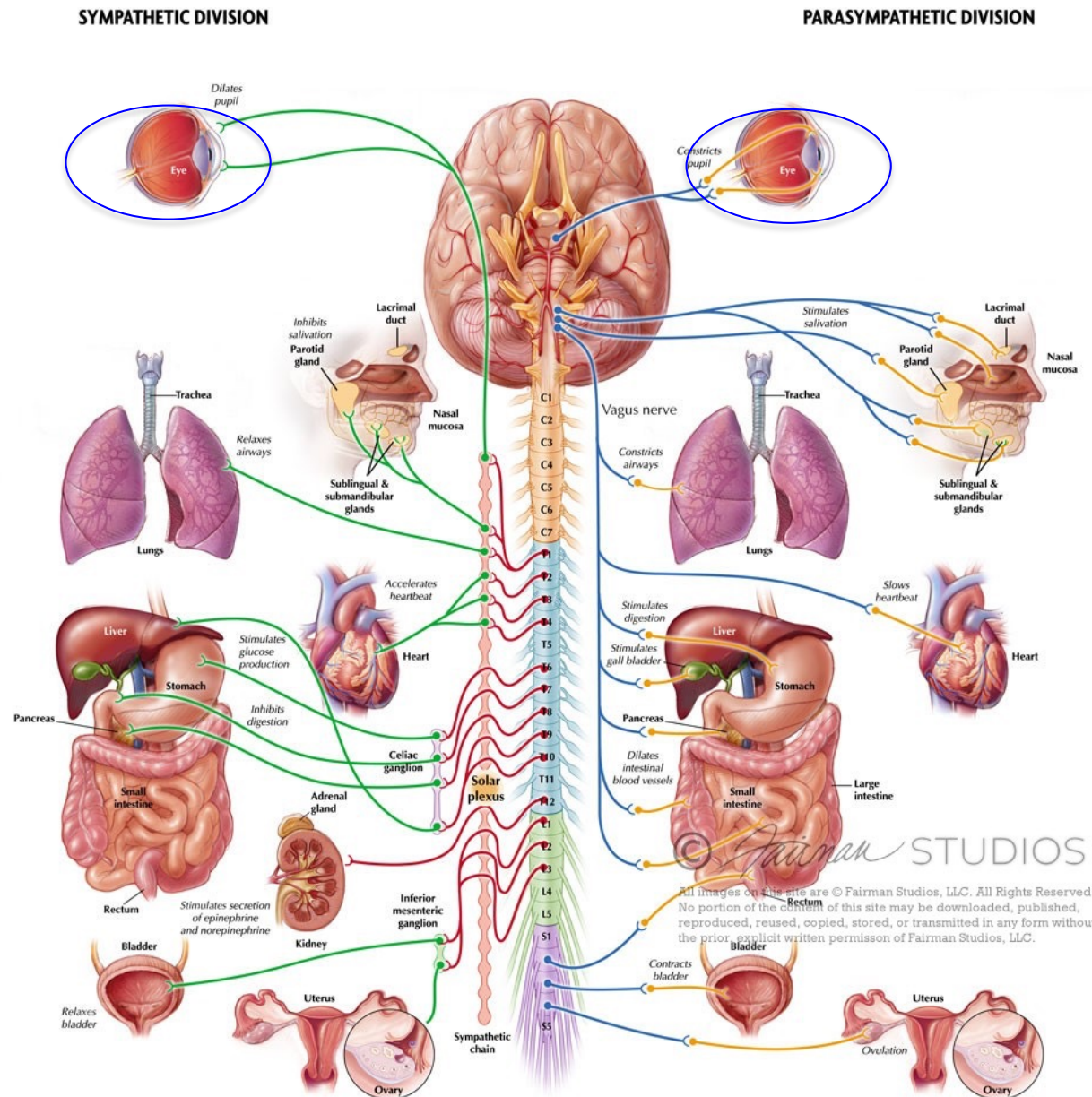
**University of Trento**

# A broad spectrum of organs are controlled by the ANS

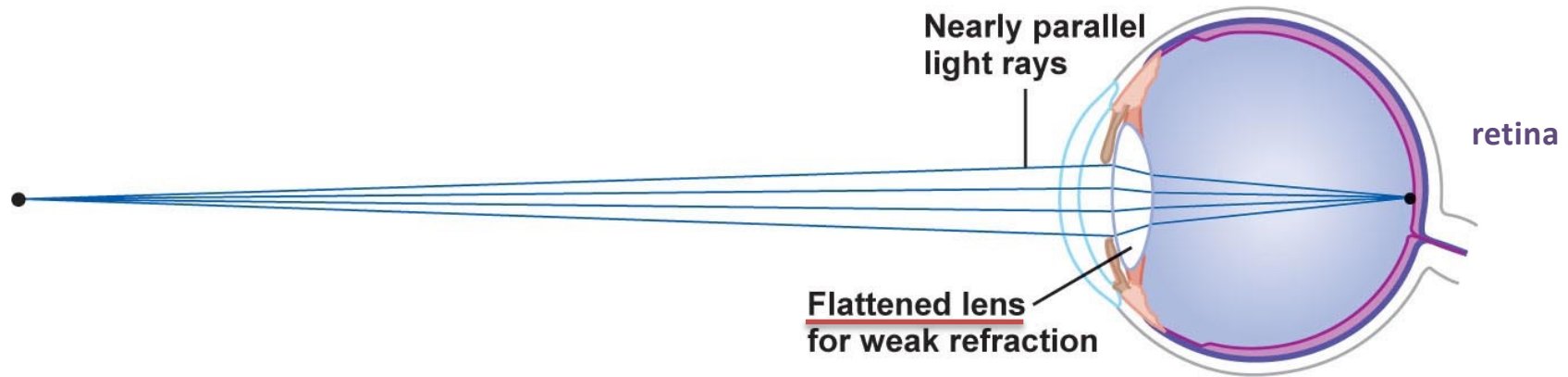


# Agenda

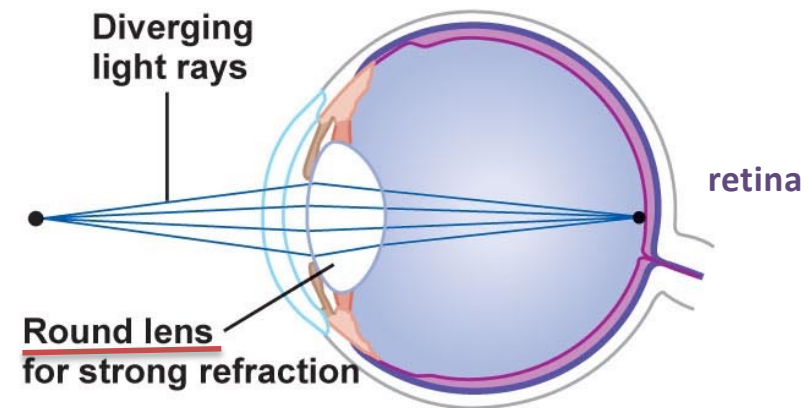
## Accommodation



# Autonomic control of Accommodation



**(a) Viewing a distant object**

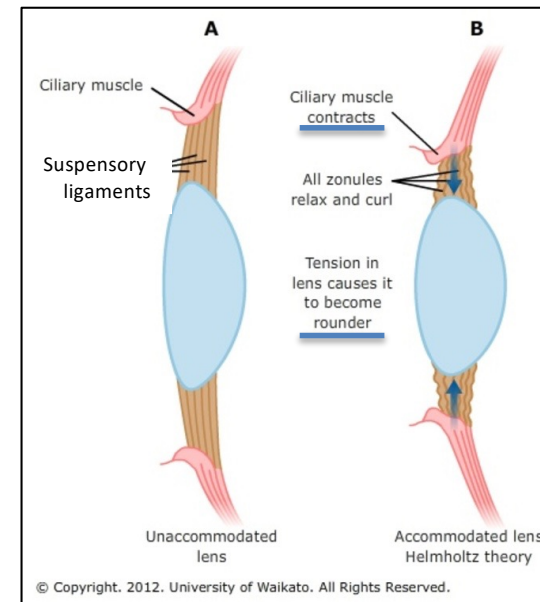
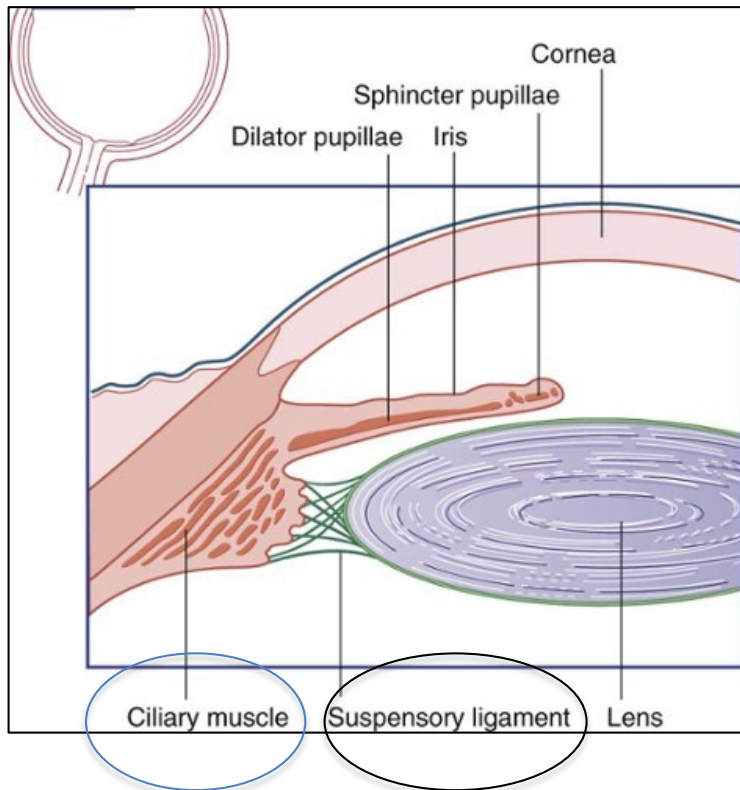


**(b) Viewing a near object**

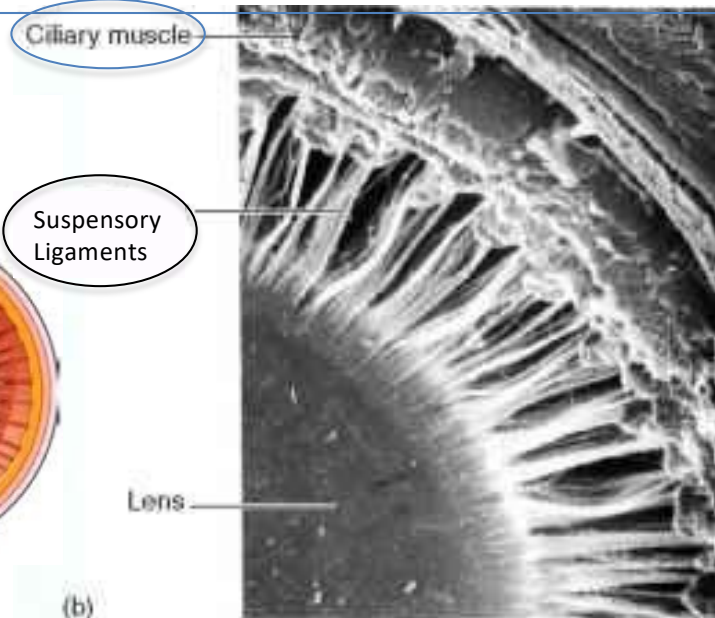
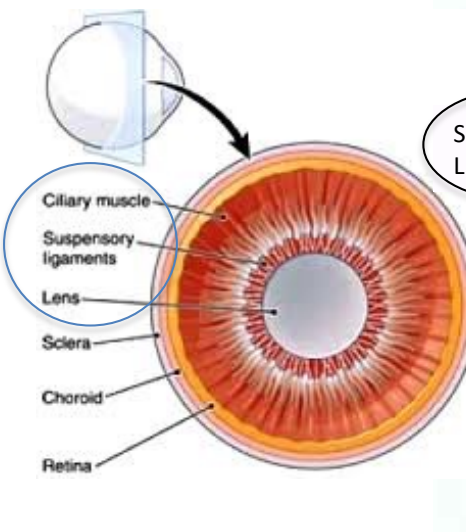
© 2011 Pearson Education, Inc.

# Smooth muscles involved in eyes accommodation (1)

## Control of lens shape

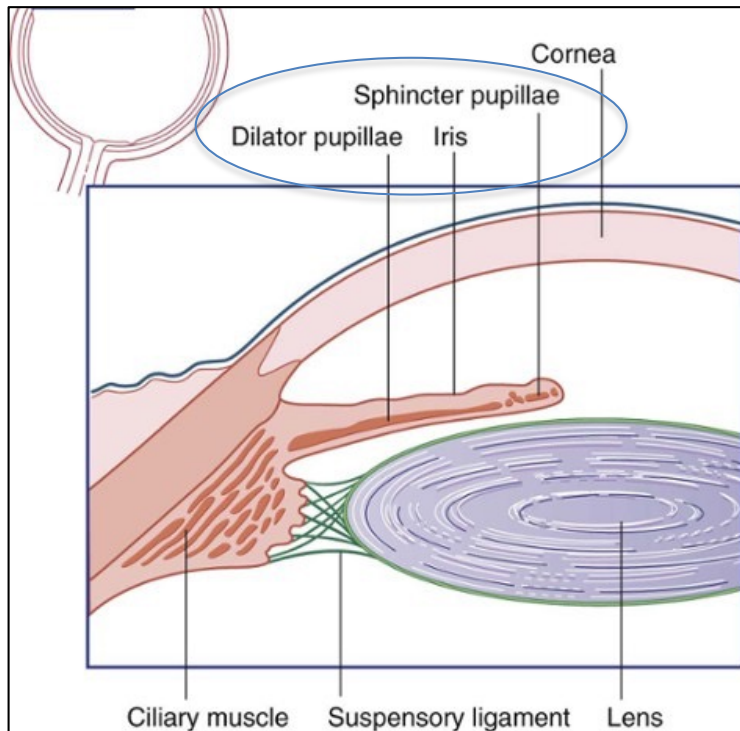


## Ciliary muscle (circular)

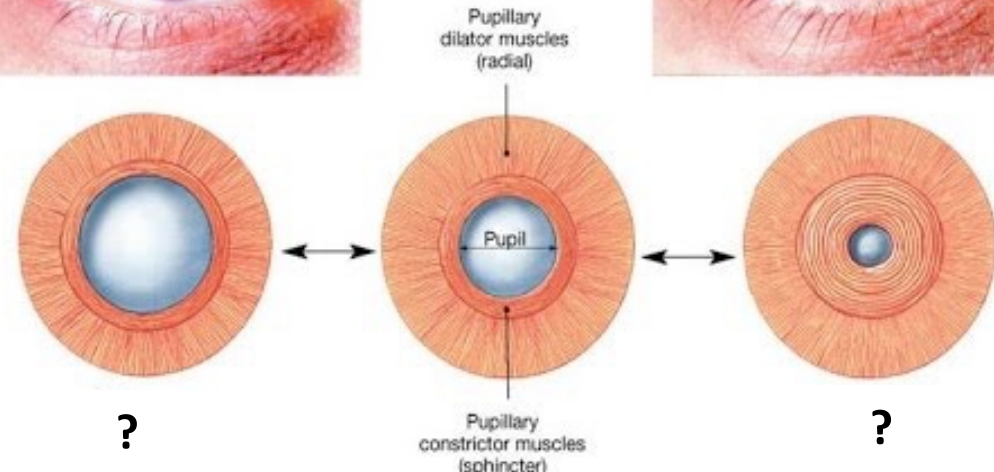
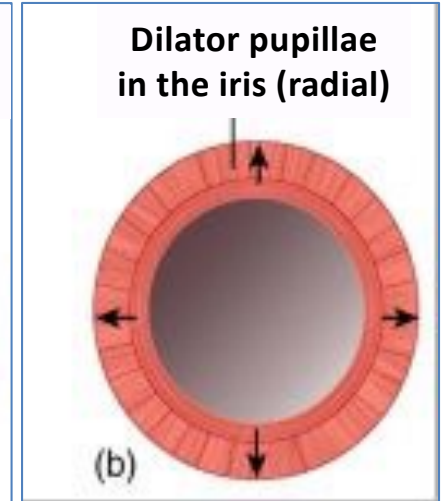
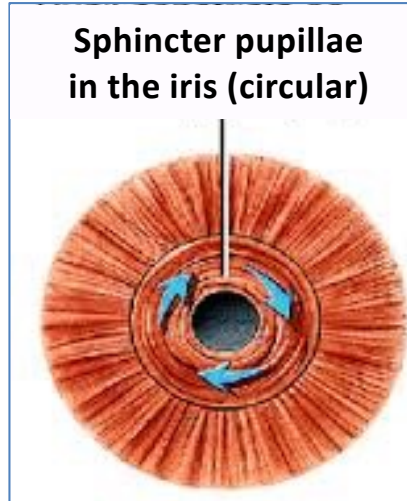


# Smooth muscles involved in eyes accommodation (2)

## Control of pupil size

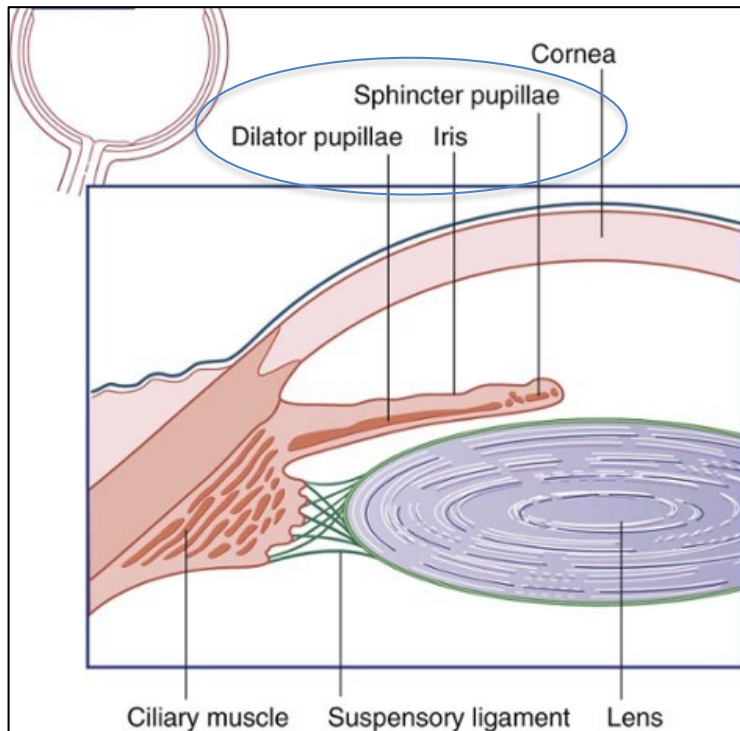


- **Response to light changes**
- Modulated during accommodation to **avoid lateral passage of light**



# Smooth muscles involved in eyes accommodation (2)

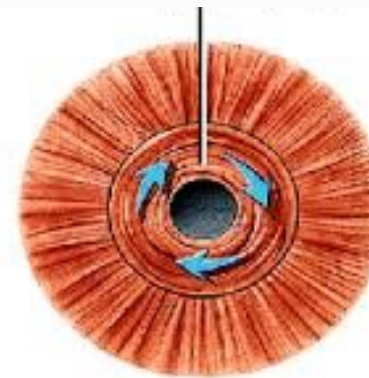
## Control of pupil size



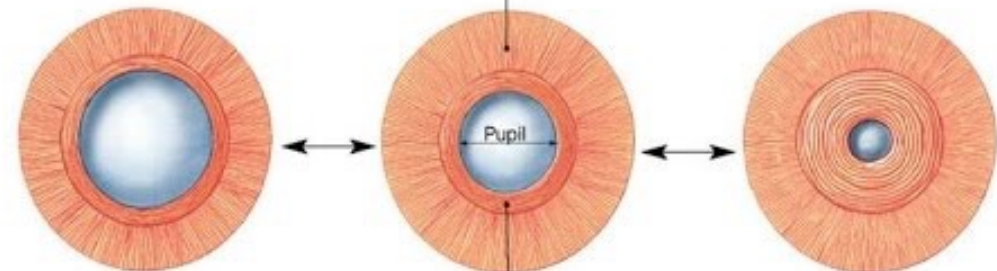
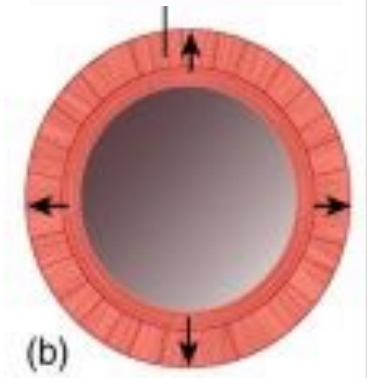
- **Response to light changes**
- Modulated during accommodation to **avoid lateral passage of light**



**Sphincter pupillae in the iris (circular)**



**Dilator pupillae in the iris (radial)**



Dilator pupillae contracts  
Sphincter pupillae relaxes

Sphincter pupillae contracts

# Autonomic control of the eyes accommodation: near vision

## Parasympathetic (Cranial III)

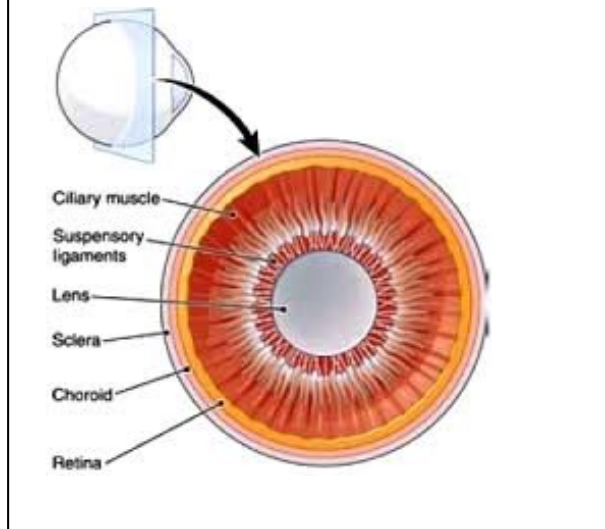
Post-ganglionic fibers induce

- **Ciliary muscle** (smooth) **contraction** which relaxes the the suspensory ligament (ticker lens)
- **Sphincter pupillae** (smooth) contractions to avoid lateral light passage

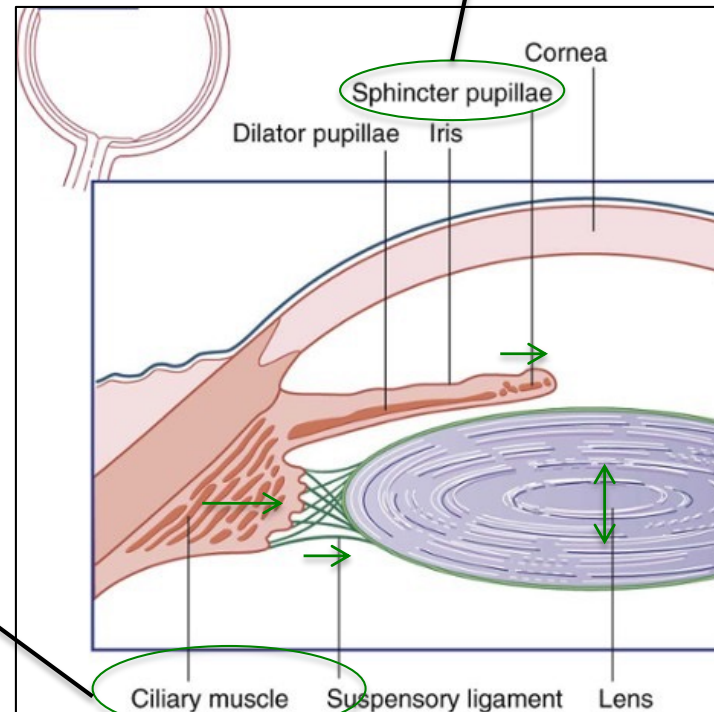
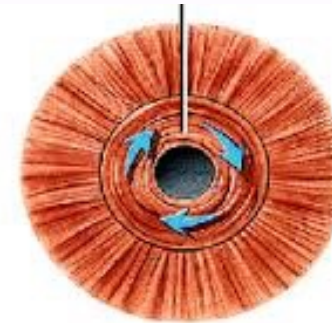
**Increased tone of medial rectus muscle** (skeletal, motorneurons) is also occurring (**convergence of eyes**)

→ **accommodation for near vision**

### Ciliary muscle (circular)



### Sphincter pupillae in the iris (circular)



# Autonomic control of the eyes accommodation: far vision

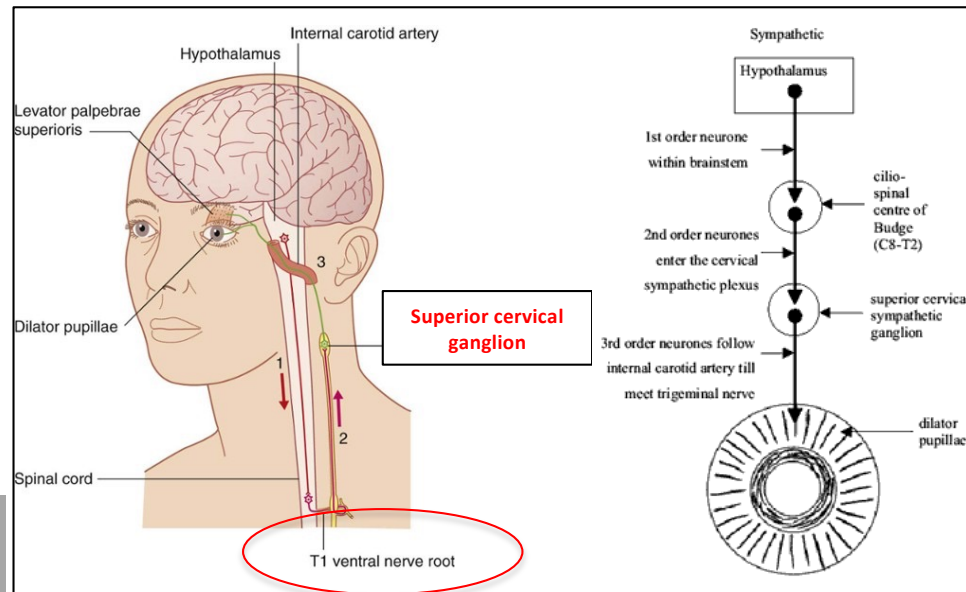
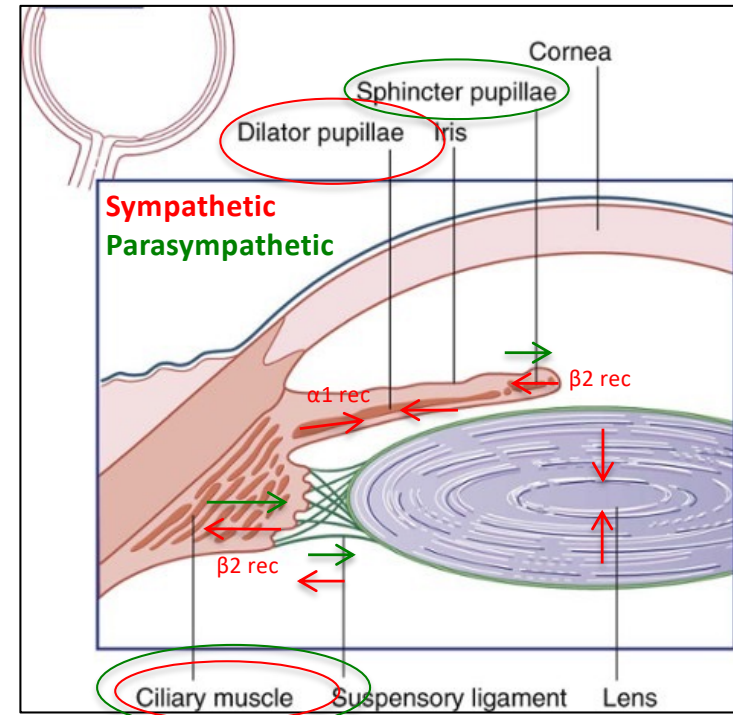
## Sympathetic (Thoracic T1-superior cervical ganglion)

Post-ganglionic fibers innervate

- **Ciliary muscle relaxation** ( $\beta_2$  receptors) contracts the suspensory ligament (flatter lens)
- **Dilator pupillae muscle contracts** ( $\alpha_1$  rec) and **sphincter pupillae relaxes** ( $\beta_2$  rec) → **dilate the pupils**

→ accommodation for far vision

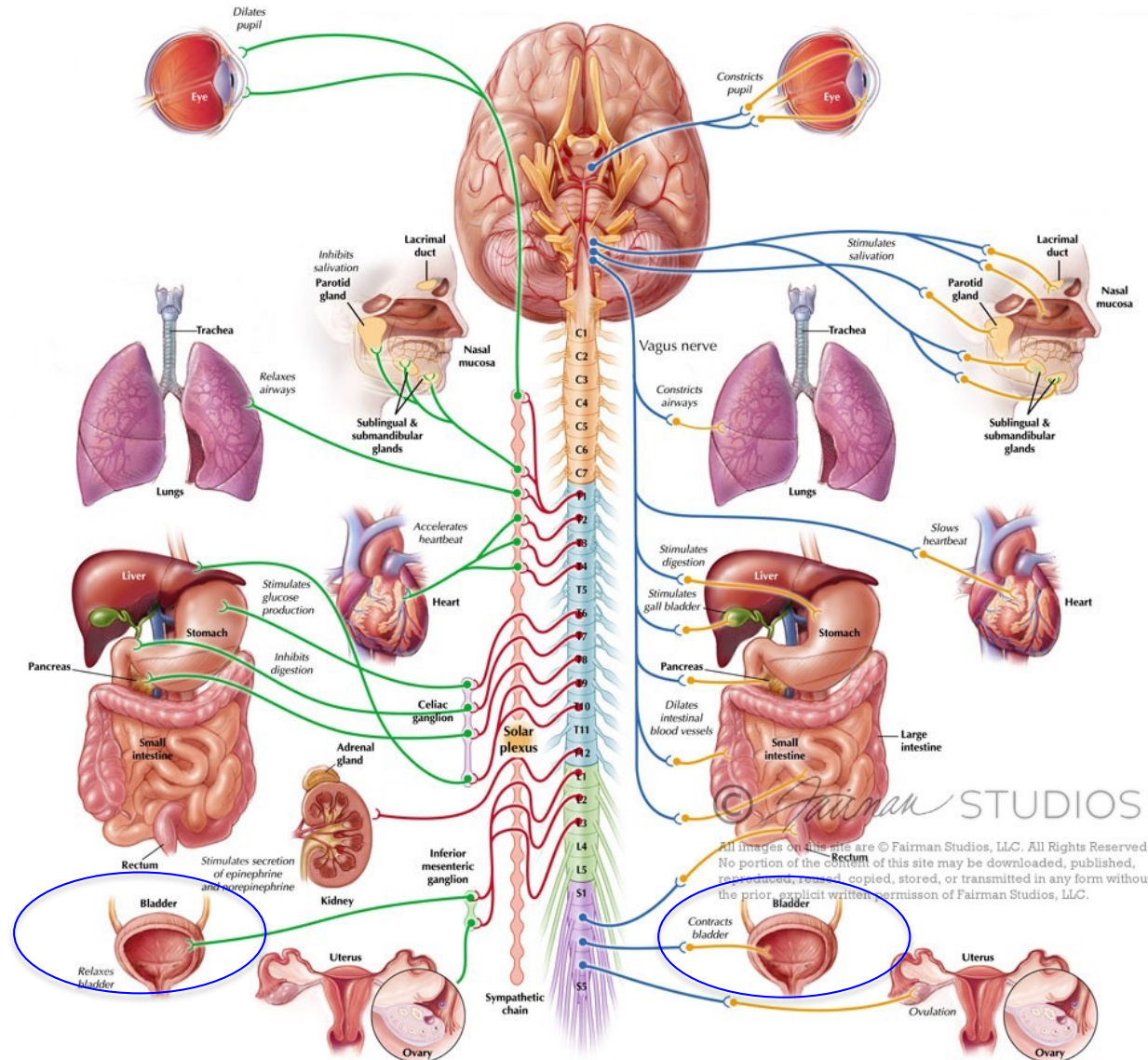
→ under stress (high sympathetic tone) could be difficult to focus close things



# Control of Micturition

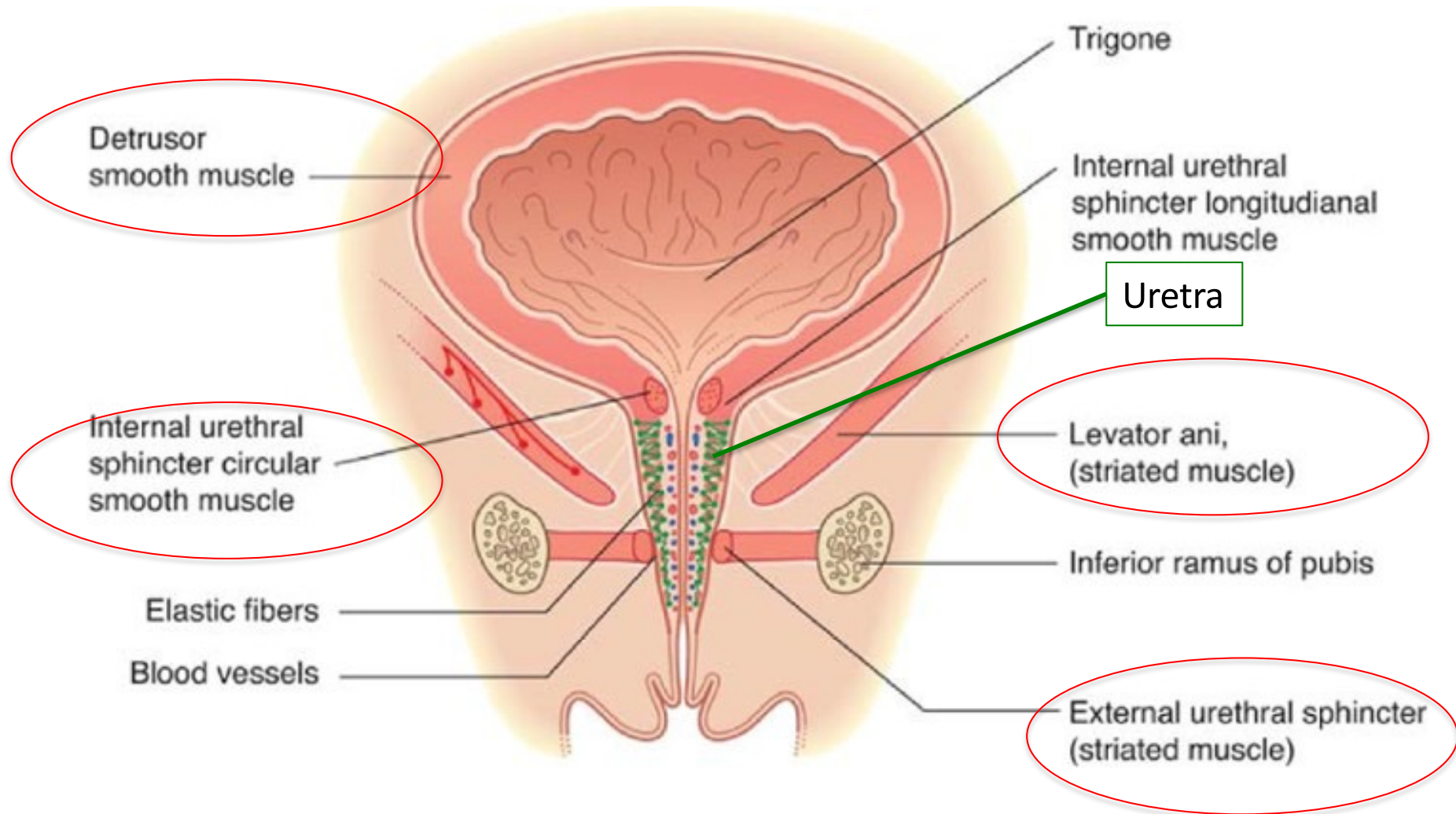
## SYMPATHETIC DIVISION

## PARASYMPATHETIC DIVISION



Bladder

# Anatomy of the Bladder

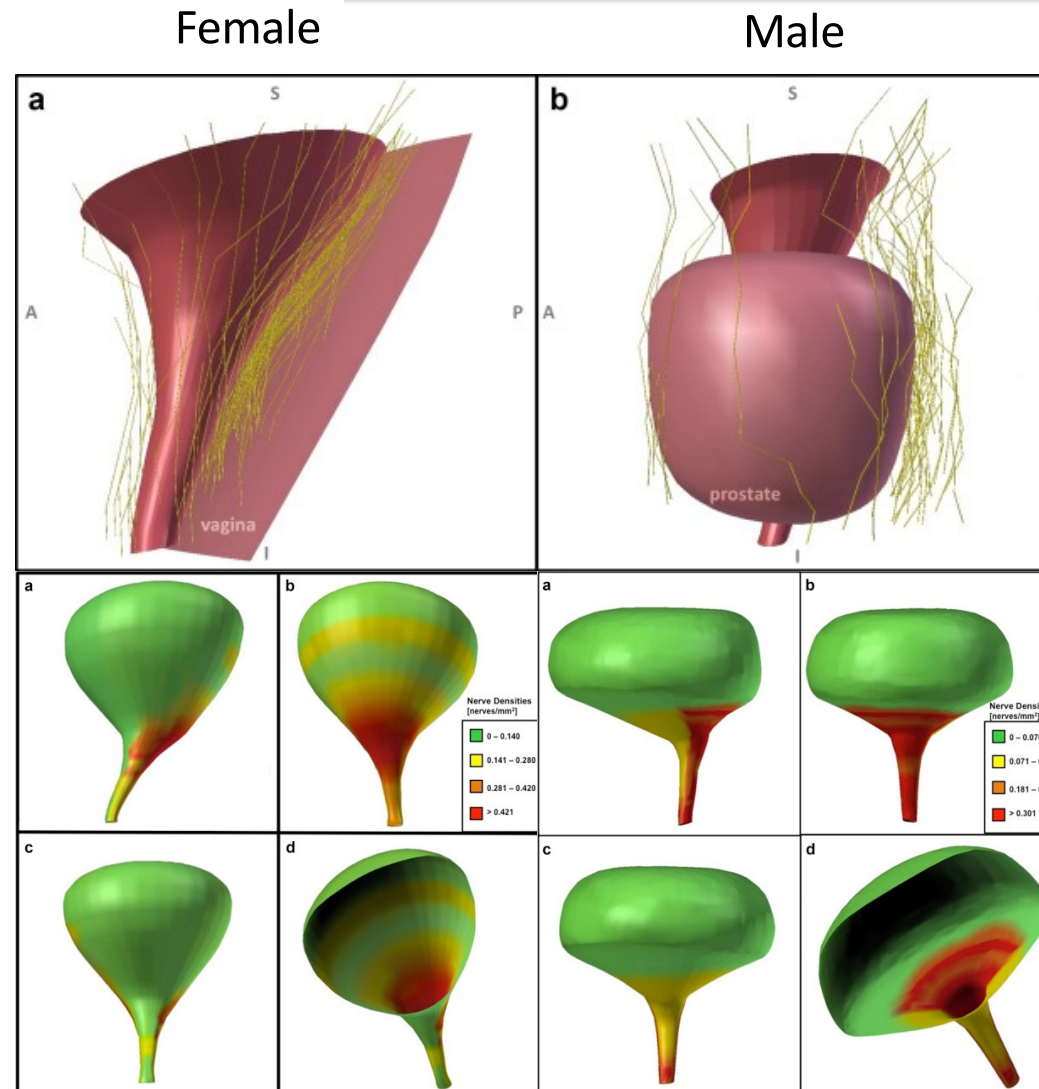


**Anatomy of the Bladder**  
(Female)

# Rich Autonomic Innervation of the bladder

3D reconstruction

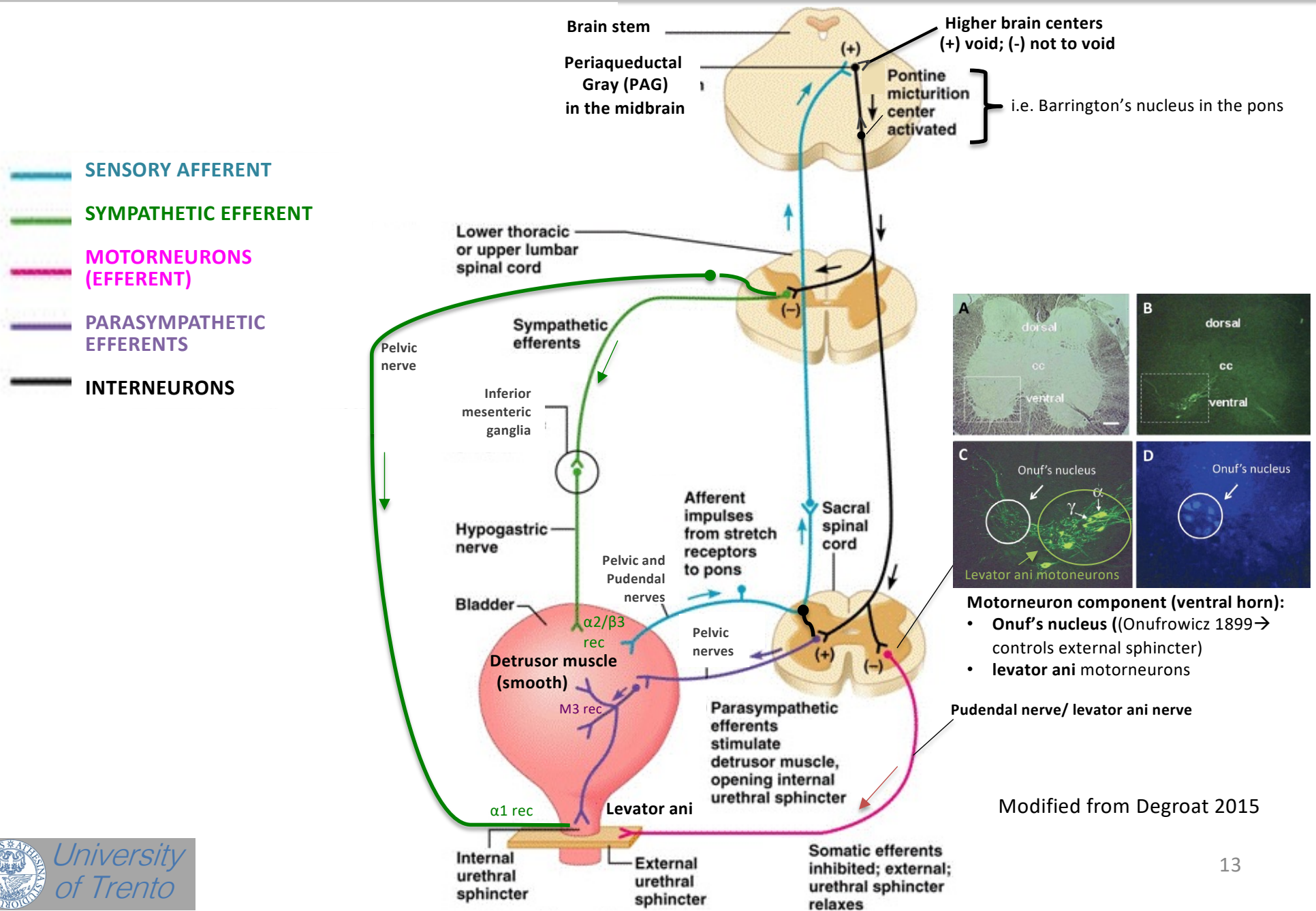
3D density maps



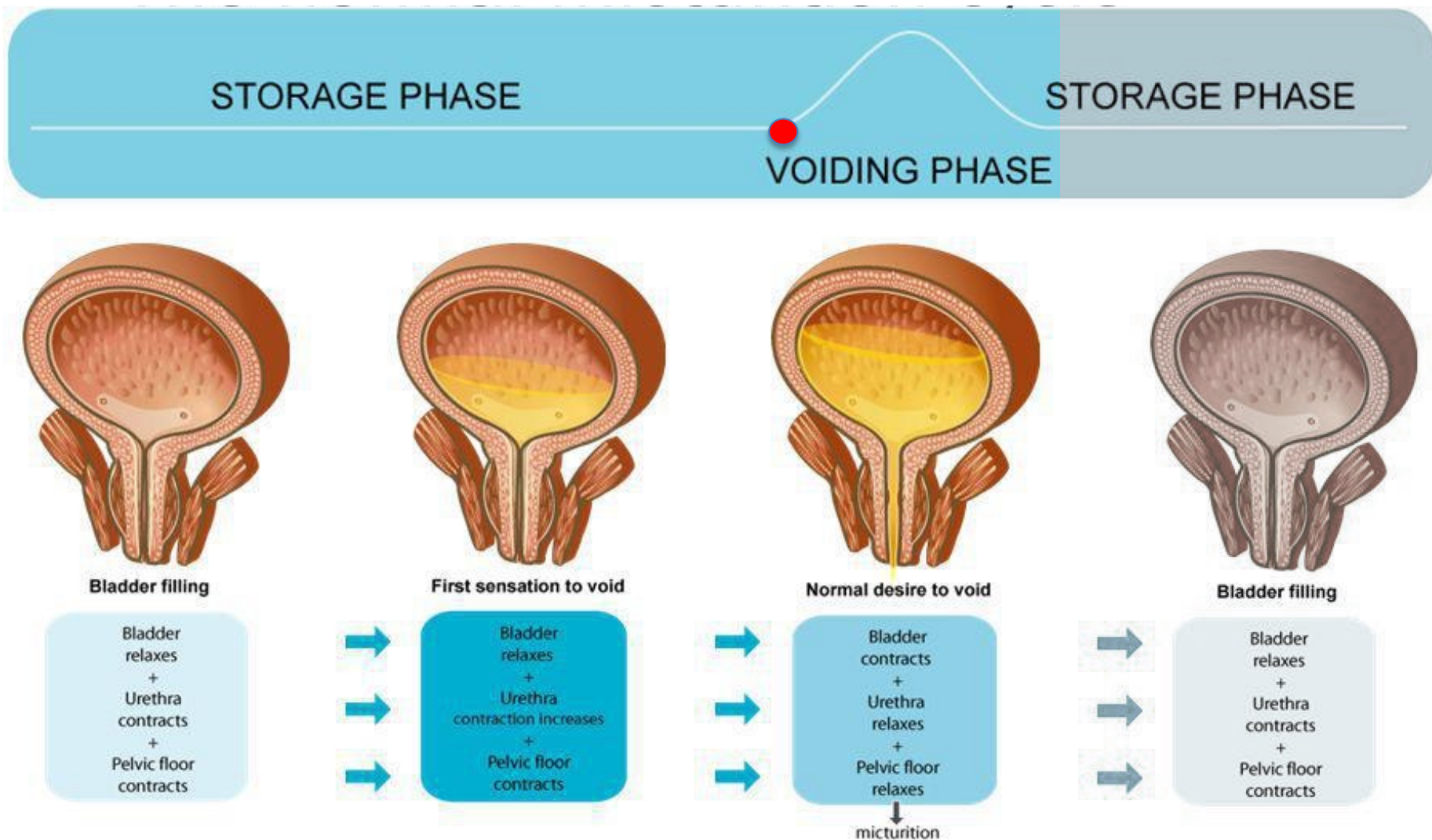
Uretra is  
richly  
innervated

**S100 antibody staining on sections (Schwann cell marker → mainly myelinated autonomic fibers)**

# Innervation of the bladder

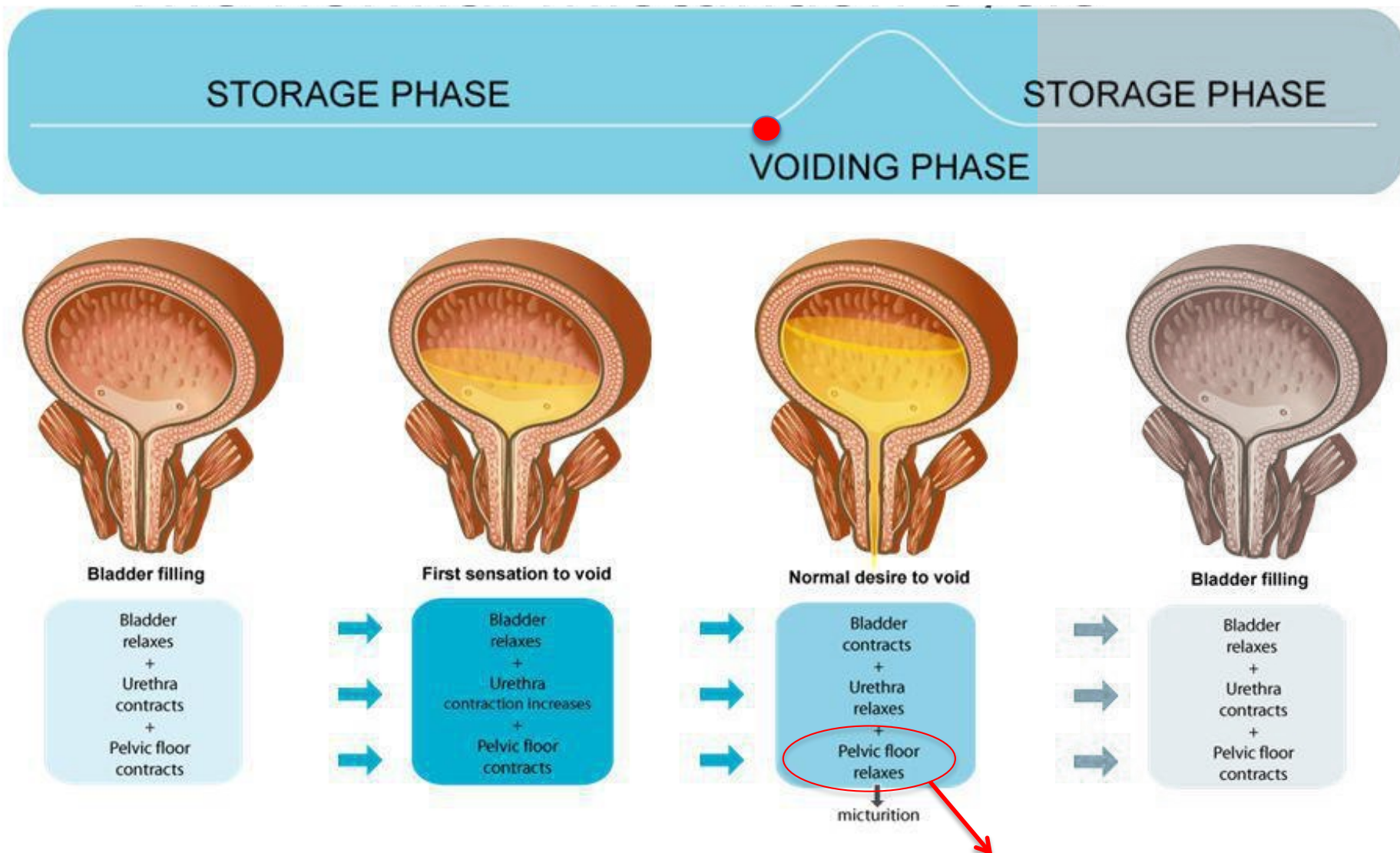


# Normal micturition cycle



Viktrup L, et al. Prim Care Update Ob/Gyns 2003;10:261-4

# Normal micturition cycle



Under voluntary control (somatic motor innervation)  
It is a learned process during development

# Neural control of micturition

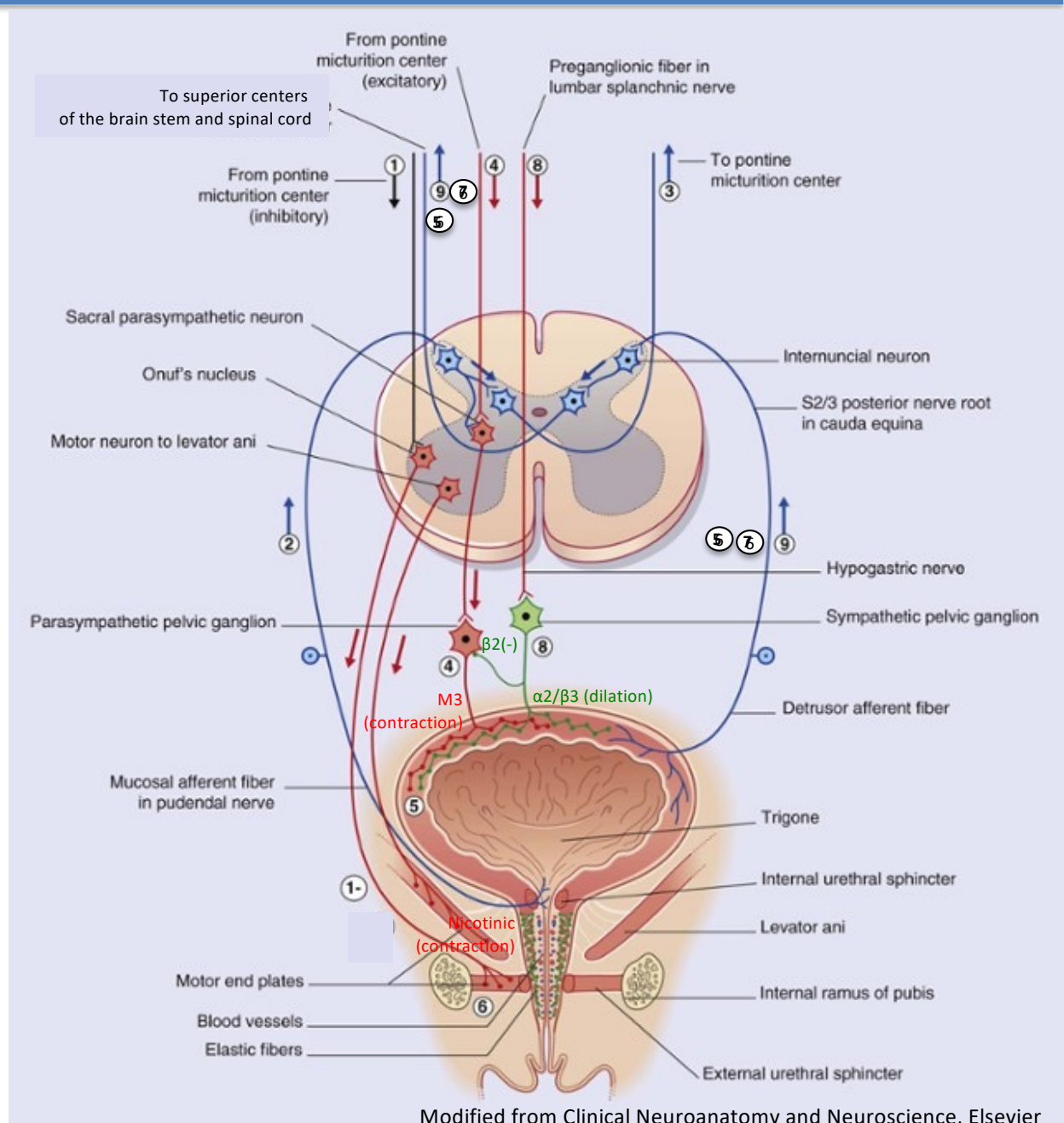
## Two phases:

- **Voiding:** steps 1→6 (parasympathetic dominates)
- **Urine storage:** steps 7→9 (sympathetic dominates)

## The somatic motor nervous system (voluntary) controls:

- **Step 1**
- **Steps 6**

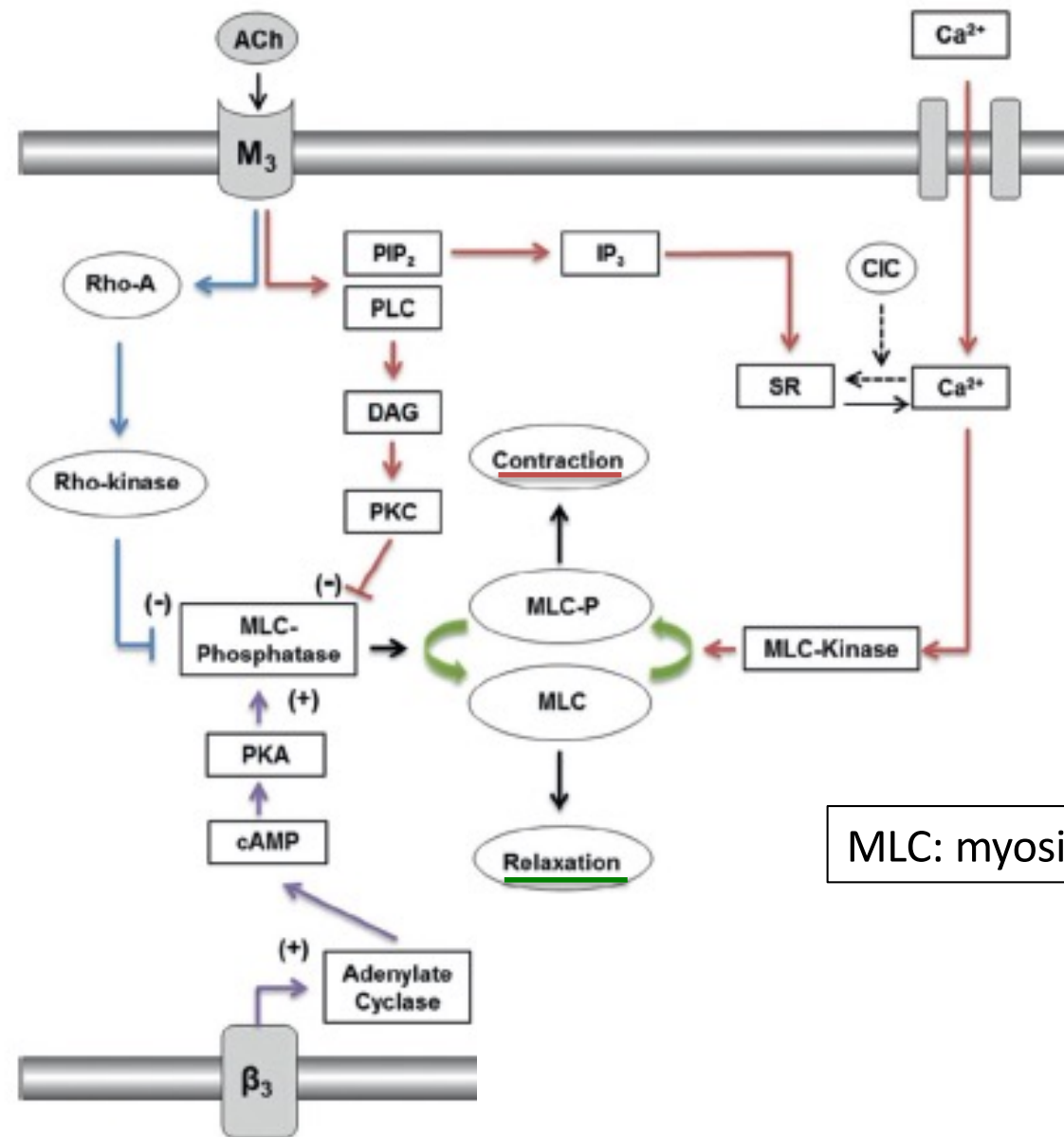
Not depicted here is an **additional sympathetic innervation** of the bladder base and **internal urethral sphincter** that are expressing **alpha 1 receptors** and therefore contracts upon NA release ensuring continence



Modified from Clinical Neuroanatomy and Neuroscience, Elsevier

# Parasympathetic vs sympathetic effects on detrusor muscle

Parasympathetic:  
contraction

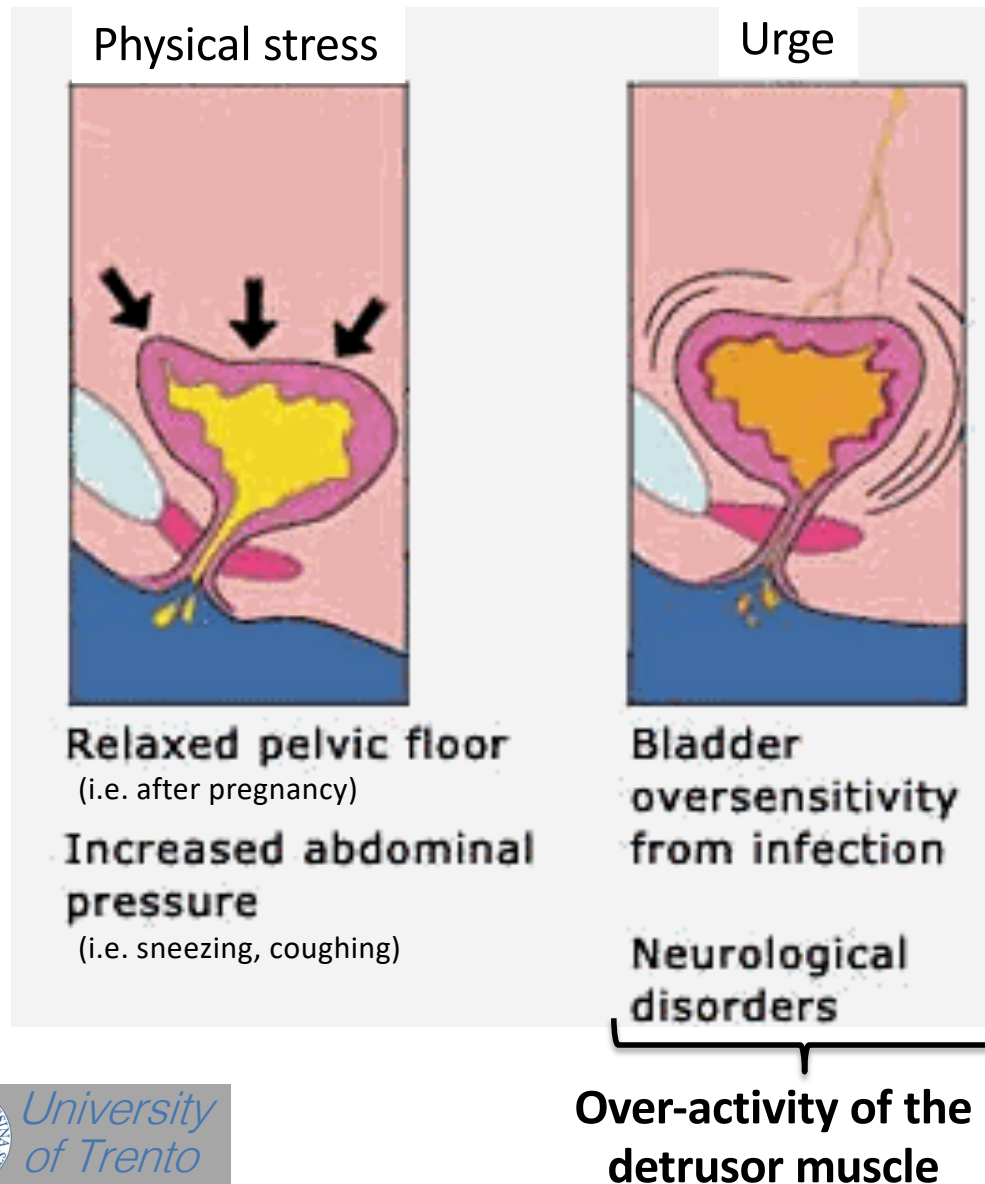


MLC: myosin light chain

Degroat 2015

# Commune types of incontinence

- **Incontinence: involuntary loss of urine**  
(30% of women are experiencing it in a moment of their life)

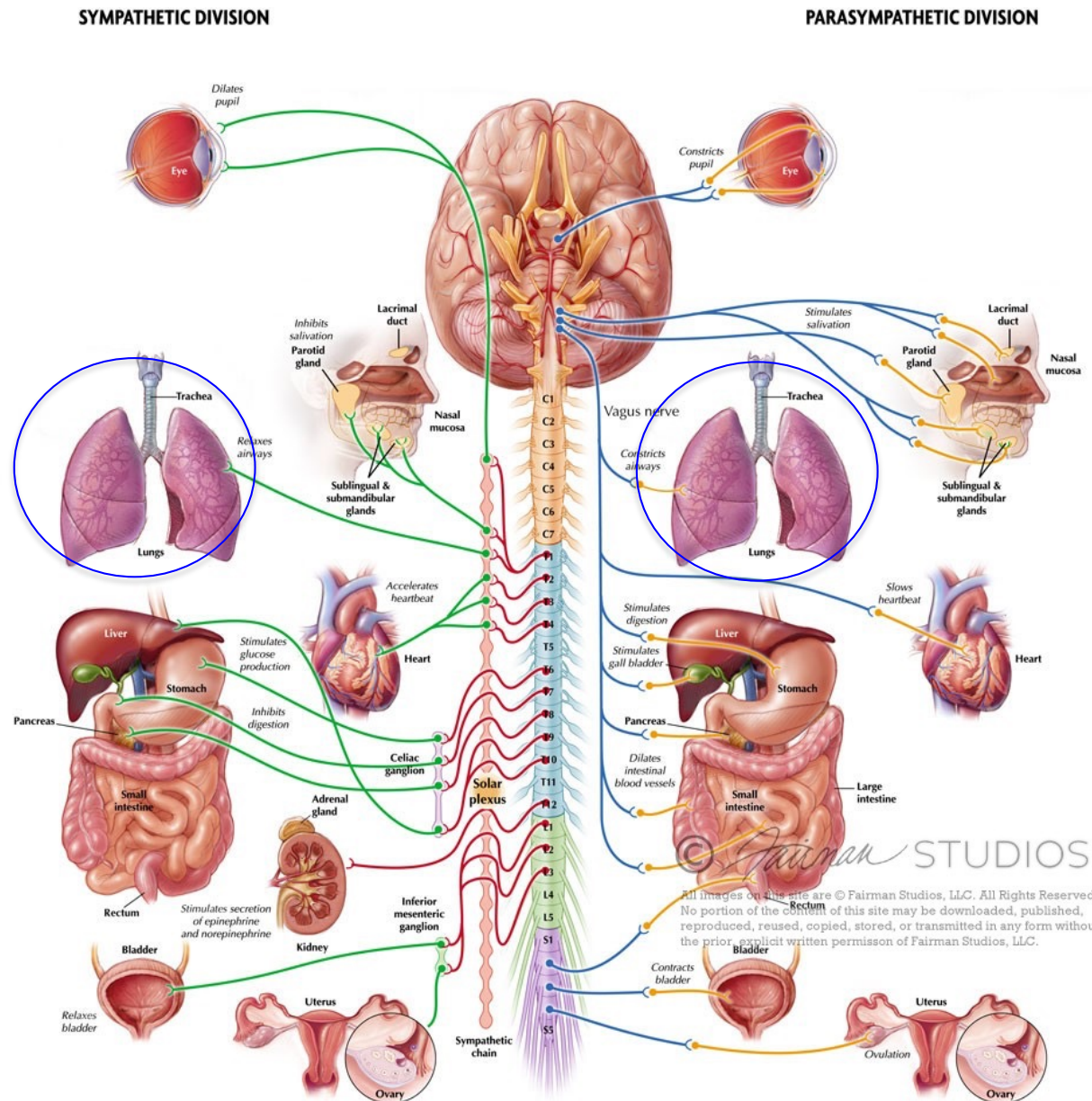


## Sensory afferent fibers types:

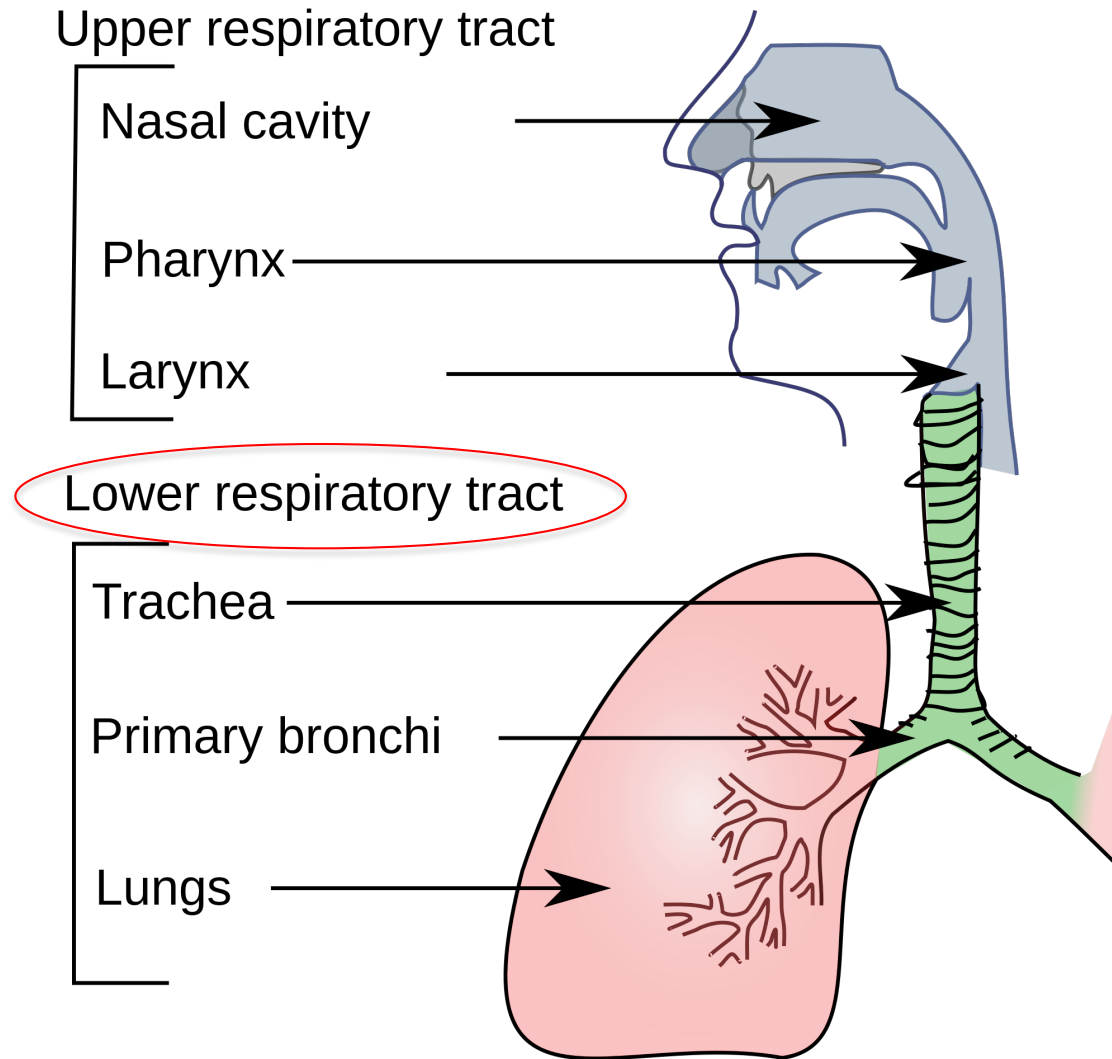
- **A $\delta$**  (thinly myelinated) respond to **passive distension** and active contraction: → **bladder tension receptor**
- **C (unmyelinated):**
  - are 2/3 of bladder fibers are **normally NOT sensitive to filling** (also called "silent" fibers)
  - are nociceptive and **involved in the "sensation" of sudden need to urinate (urgency)** and possibly **involved in the pathogenesis of Overactive Bladder (OAB)**

# Control of airways mucous secretion

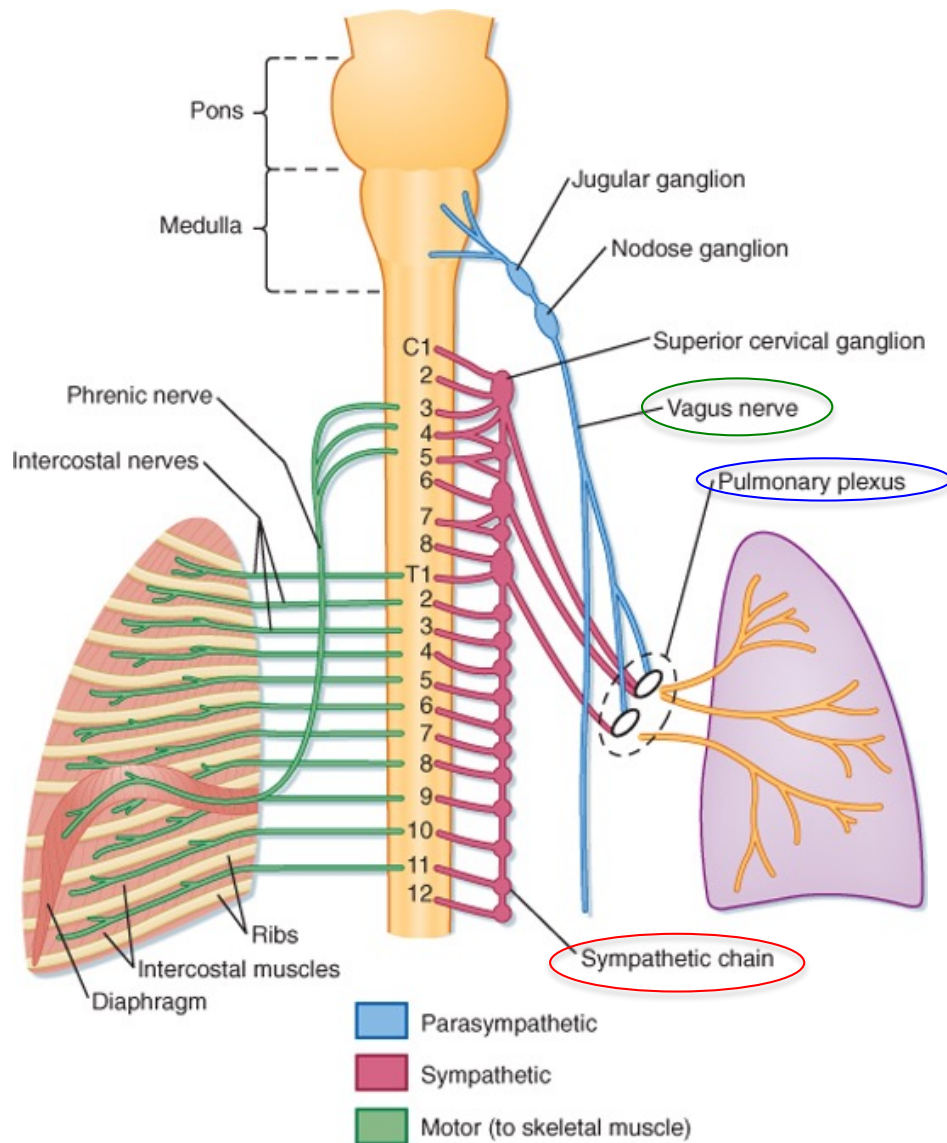
## Airways



# Lower respiratory tract



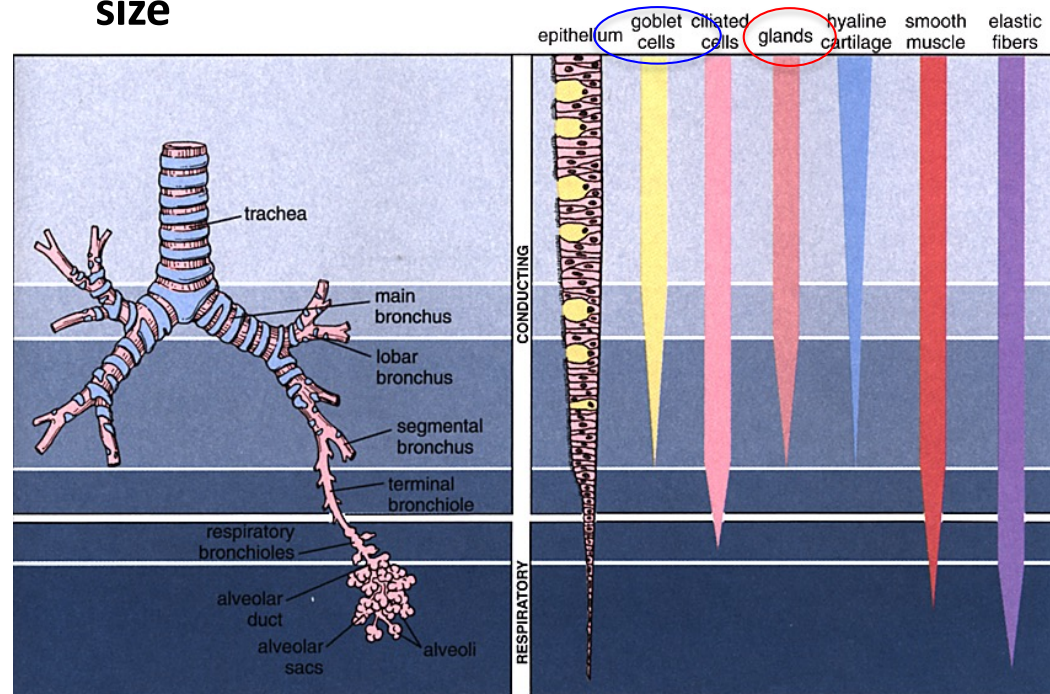
# Autonomic Innervation of the Lower airways



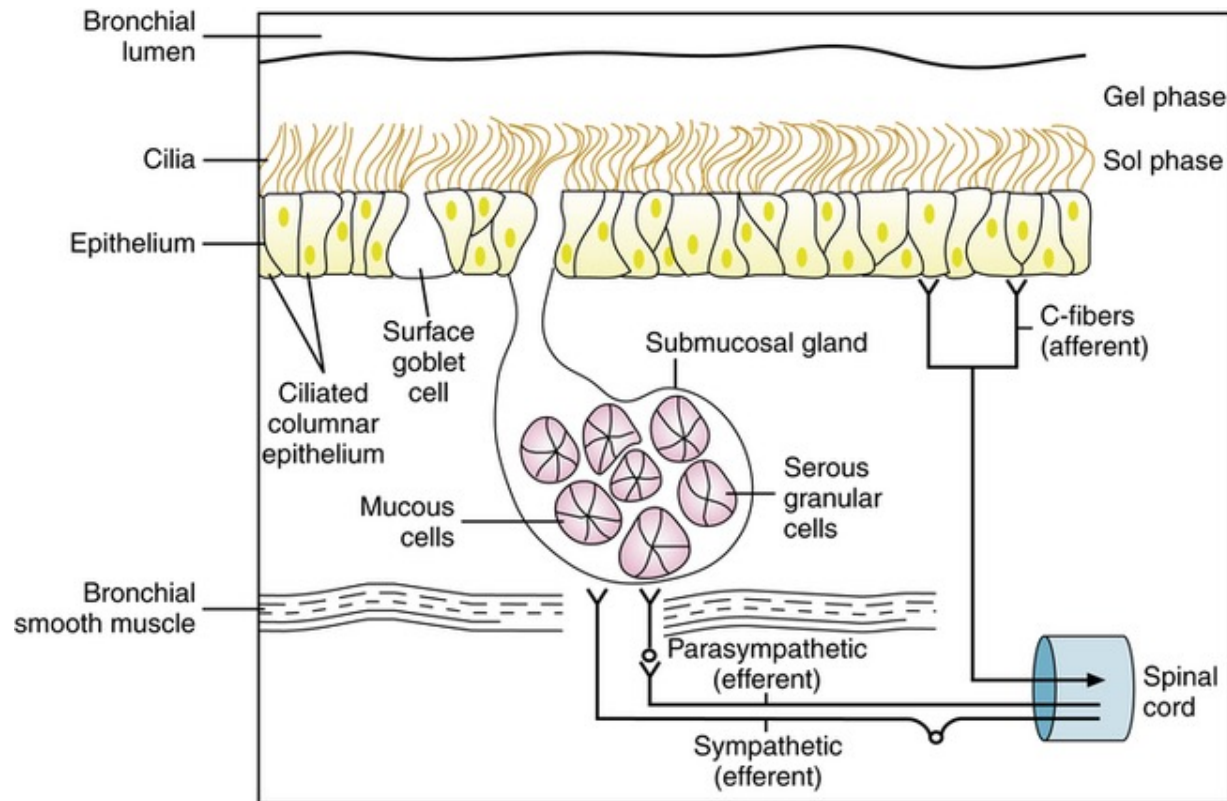
# Airways secretion



- Mucous secretions provide a **physical barrier to airborne pathogens** and contribute to their destruction by containing **antimicrobial chemicals** (i.e. Lysozyme)
- Submucosal gland outnumber goblet cells (20:1)
- Glands decrease with the **decrease of airways size**



# Innervation



Glands are innervated by the **parasympathetic system** and **sympathetic**  
**→all are promoting increase in mucous secretion**

# Summary

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- **Accommodation (near or far vision)** is mediated by a tight **control of smooth muscles** by the **parasympathetic and sympathetic** nervous systems
- The **iris is containing circular and radial muscles** that when contracted respectively by **parasympathetic and sympathetic systems** are **determining opposite effects on the pupil size**
- The **parasympathetic NS** predominate in **near vision**, the **SNS** in **far vision**
- **Micturition** is controlled by the **coordinated action** of somatic **motor, sensory, parasympathetic, sympathetic** innervation
- **Voiding** may **initiated voluntarily** by inhibiting the contraction of the external sphincter muscle
- The **micturition cycle** is involving concerted **brainstem and spinal reflexes**
- Both the **parasympathetic and sympathetic branches** are contributing to stimulation of **secretions** from glands at the level of the **airways**

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Elsevier Saunders

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**ISBN-10:** 0702037389

**Chapters:** 13, 23

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Buijs & Swaab; Elsevier

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